

Assessment task — Investigating changing states of matter**Task**

Plan, conduct, and evaluate an investigation exploring a variable that affects changes in the states of matter using ice, and demonstrate understanding of the physical properties of solids, liquids, and gases.

Instructions

Part A: Understanding solids, liquids and gases

Classify the properties of solids, liquids and gases.

Part B: Investigating changing states of matter

Plan, conduct and evaluate an investigation of a variable that affects the amount of time it takes for an ice cube to melt.

Conditions

The assessment task is to be completed on these pages and other blank paper where required. It is carried out under test conditions. Read and respond to the questions carefully.

Part A: Classifying substances

1. These solids, liquids and gases are all mixed up. Draw a line or highlight which of the materials are solids, liquids and gases.

Orange Juice

Birthday Candle

Wooden box

Liquid

Rice

Silver

Helium

Solid

Flour

Olive oil

Golden Syrup

Ice block

Gas

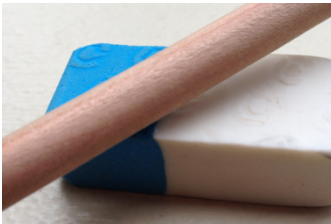
Air

2. Fill in the table by putting a cross (x) in the correct box or boxes.

	SOLID	LIQUID	GAS
a. It fills the shape of its container			
b. It stays the same shape			
c. The air around us is made of this			
d. If you freeze a liquid it will become a			
e. If you boil water it will become a ...			
f. It has mass			

3.

- a) Classify each of the objects below as a solid, a liquid or a gas.
- b) Identify physical properties of solids, liquids and gases.

Object	Solid, liquid, gas?	Identify physical properties	Particles Draw a Diagram
(i) Honey 			
(ii) An eraser 			
(iii) Bubbles in soft drink 			
(iv) Sand 			

4. Use the words from the list below to complete the sentences:

Word list: **heat, solids, freezes, shape, volume, cool, melts**

- a. Liquids change _____ when you move them in a container.
- b. _____ don't change shape when you move them.
- c. Moving a liquid from a bottle to a glass does not change its _____ .
- d. When a solid _____ it turns into a liquid.
- e. When a liquid _____ it turns into a solid.
- f. To turn a solid into a liquid, you must _____ it.
- g. To turn a liquid into a solid, you must _____ it.

Part B: What affects how quickly ice melts?

You will **plan, conduct, and evaluate** a scientific investigation to test how a **chosen variable** affects the **time it takes for an ice cube to melt**.

Materials and Equipment

- 3 ice cubes (same size)
 - 3 plates, cups, or containers (each made of different materials or conditions)
 - Stopwatch or timer
 - Recording table or notebook
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Method

 **Note:** You will be timing how long it takes each ice cube to melt completely.

1. Collect all your materials and set them up in a safe space.
2. Place one ice cube in the first container or on the first surface.
3. Start the stopwatch and time how long it takes for the ice cube to melt completely.
4. Record the result in your data table.
5. Repeat steps 2–4 with a second ice cube, using a different container, surface, or condition.
6. Repeat steps 2–4 again with the third ice cube, using a third variation.
7. Ensure all other conditions (like room temperature and ice cube size) remain the same.
8. Record all three times in the results table.

Plan

Prediction (Hypothesis)

What do you think will happen? Use science words like *melt*, *faster*, *slower*, *heat*, or *surface*.

 I predict:

Because:

Variables

 **Independent variable** (What I will change):

 **Dependent variable** (What I will measure):

 **Controlled variables** (What I will keep the same):

Safety and Intercultural Considerations

 **Safety risks:**

☐ Spills on the floor

☐ Cold hands from holding ice

☐ Slippery surfaces

Other: _____

 **What I will do to stay safe:**

Record Your Data

Use the table below:

Surface/ Condition	Time taken to melt (minutes)	Observations

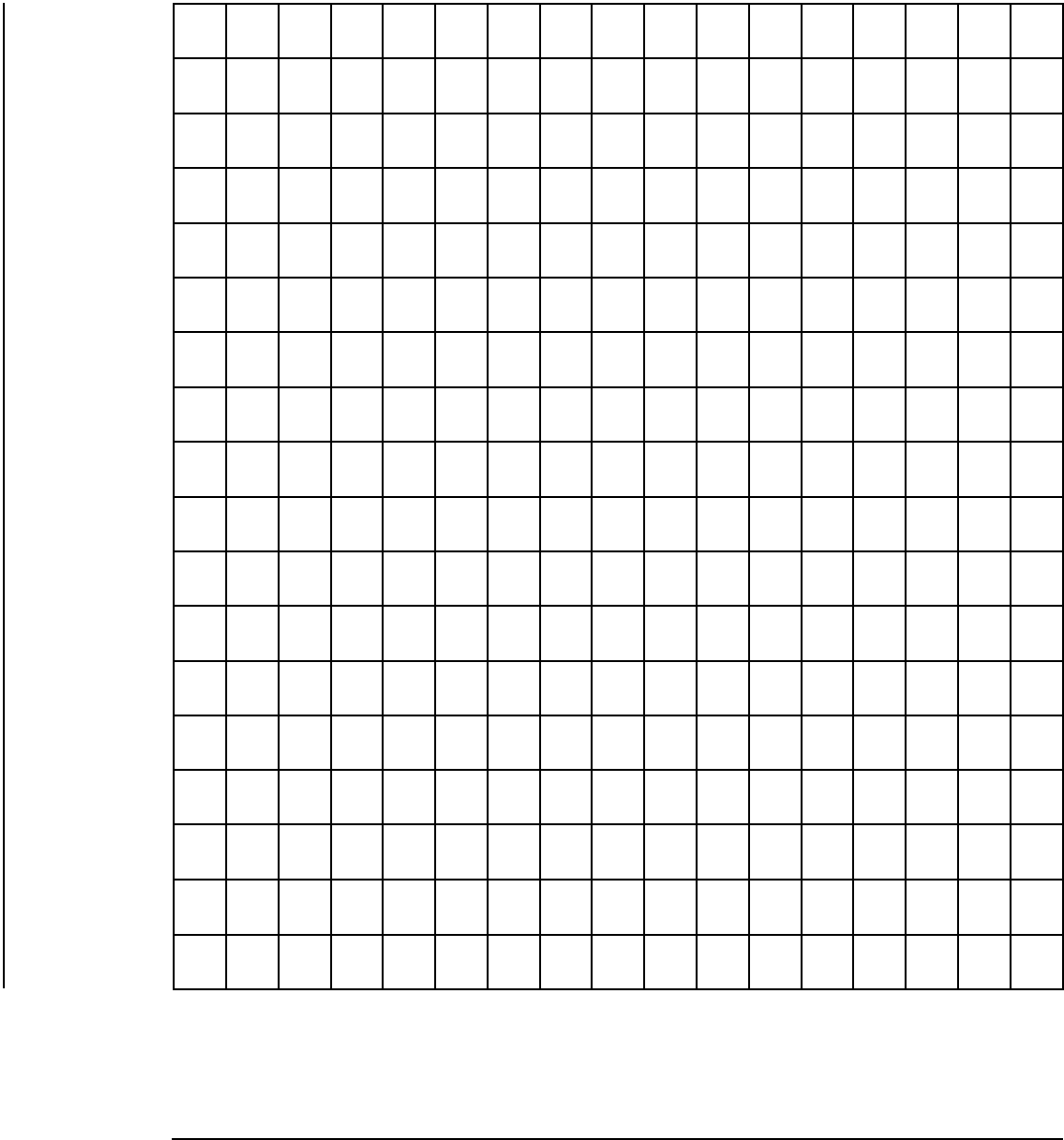
 Timer used? ☐ Yes ☐ No

 Photos taken? ☐ Yes ☐ No

Data Representation

Use the space below to draw a **bar graph** to show your results.
(Label your axes!)

 **Graph Title:** _____



What Did You Notice? (Patterns/Trends)

 I noticed that:

The fastest melting condition was: _____

The slowest melting condition was: _____

Evaluate Your Investigation

Was this test fair? Yes/No

a) Describe why or why not.

b) What changes would you make next time to improve the fairness of your investigation?

c) Describe how these changes would improve fairness.

Conclusion

 My investigation showed that:

My prediction was: ☐ correct ☐ incorrect (tick one)

What's Next?

What new question could you investigate next time?

 I wonder what would happen if...
